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FAQ/General

	• FAQ/General [FrontPage] [TitleIndex] [WordIndex]

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Join us on the CentOS-Docs mailing list to discuss this, and other documentation work!

Frequently Asked Questions about CentOS in general

Contents

1. What is CentOS Linux?
2. How long after Red Hat publishes a fix does it take for CentOS to publish a fix?
3. Where are the Release Notes for a given version?
4. How do I get Updates?
5. Many RPMs still contain the name redhat, rhel, or rh. Shouldn't these be changed?
6. Does CentOS change the upstream Source RPMs?
7. Is there a Contrib area?
8. What is CentOS's relationship with Red Hat®, Inc. or RHEL?
9. Where can I get package XyZ.rpm for CentOS?
10. What architectures are supported?

11. The upstream provider offers Enterprise Linux in several flavors, AS, ES, WS, PWS, etc. Which one is CentOS like?
12. How can I easily compare what major package versions are in each version of CentOS?
13. What are all the CentOS repositories (directories) and what is each one for?
14. How do I download and burn the CentOS isos?
15. Can I add X to my server after install? Or I installed from the Server ISO and it didn't install X, how do I install X?
16. How do I create a xorg.conf file?
17. What are the Maximum number CPUs, Maximum filesystem size, Minimum / Maximum Memory and other specifications associated with the different CentOS versions?
18. There is no mp3 support in CentOS! Or is there? How about other multimedia formats?
19. What is the support "end of life" for each CentOS release?
20. Where can I get the latest version of XyZ.rpm for CentOS? I cannot find it anywhere.
21. A PCI audit says I am running a version which has CVE exploits in it
22. How do I install or update on a system with no network connection?
23. I installed the x86_64 version, so why do I have i386 packages, and can I get rid of them?
24. How do I provide appropriate information about my system when asking questions?
25. How do I upgrade from one major release to another?
26. Why was there no package selection step when I've tried to install CentOS using the LiveCD/LiveDVD?
27. Why is the CentOS Project Spamming me?
28. Why is the CentOS Project taking over my Website?
29. How does CentOS versioning work?

1. What is CentOS Linux?

CentOS Linux provides a free and open source computing platform to anyone who wishes to use it. CentOS Linux releases are built from publicly available open source source code provided by Red Hat, Inc for Red Hat Enterprise Linux. This source code is available on the CentOS Git website.

CentOS Linux is the Community Development Platform for the Red Hat family of Linux distributions. See this FAQ for more information.

CentOS conforms fully with Red Hat, Inc's redistribution policies and aims to be functionally compatible with Red Hat Enterprise Linux. CentOS mainly changes packages to remove trademarked vendor branding and artwork.

CentOS Linux does not contain Red Hat Enterprise Linux or Fedora Linux; nor does it have any of their certifications, although it is built from the same source code as the Red Hat Enterprise Linux.

2. How long after Red Hat publishes a fix does it take for CentOS to publish a fix?

Our goal is to have individual RPM packages available on the mirrors within 72 hours of their release, and normally they are available within 24 hours. Occasionally packages are delayed for various reasons. On rare occasions packages may be built and pushed to the mirrors but not available via yum. (This is because yum-arch has not been run on the master mirror. This may happen when issues with upstream packages are discovered shortly after their release, and if releasing the package would break it's functionality.)

Update Sets (see this FAQ) will have Security Errata released was stated above, while the BugFix and Enhancement errata are actually tested more rigorously and released after the new ISO for the Update Set is produced. The goal for release of a new point release or update set is four to eight weeks after the release by upstream. For more details see the explanation of The CentOS Rebuild and Release Process

During the period between the upstream release of a new minor version and the CentOS release intermediate updates will be published in a Continuous Release (CR) repository.

3. Where are the Release Notes for a given version?

Release Notes have last minute documentation or describe a commonly encountered problem, when a given release has a variation from the formal documentation. Release Notes are also used when there is a need to emphasize some change. The Release Notes vary, not unexpectedly, both by major release (CentOS 8, 7, ...), and also with each new minor 'point' update within a major release. This compilation collects links to each Release Note.

4. How do I get Updates?

CentOS ships with an application called yum (in CentOS Linux 7) and dnf (in CentOS Linux 8 and CentOS Stream) that serves as our tool for performing updates and package installation. Please see PackageManagement, PackageManagement/Yum, and **man yum** or **man dnf**.

There is a separate FAQ question concerning how to import the CentOS RPM signature key.

5. Many RPMs still contain the name redhat, rhel, or rh. Shouldn't these be changed?

The upstream vendor is using open source (mostly GPL) software in their business model. They take software that other people write (Gnome.org, X.org, KDE.org, OpenOffice.org to name a few). They repackage the source files into RPM format for redistribution. Because they chose an open source model to obtain the software they distribute, they must provide their source code to others. That is how the GPL works. The upstream vendor provides much added value by creating the Source RPMS and distributing them. They also fix problems in software and provide feedback to the software developers ... this is what makes open source software work.

The CentOS Project takes the publicly available source packages (SRPMS) provided by the upstream vendor and creates binary (installable) packages for use by anyone who wishes to use them.

Some packages contain Trademark information and the upstream vendor has specifically created a guide to redistribute software built from their publicly available sources. You can read about it here. We support the upstream vendors Trademark rights and strive to be in full compliance with those guidelines.

The CentOS project is not interested in taking credit for work done by others, so where possible we will leave all vendor file names as they are. If we must make any changes to a package (due to trademark restrictions, to setup a configuration file, etc.), it will have a .centosx in the filename (the x is the CentOS version ... 5 for CentOS-5.x, 6 for CentOS-6.x, etc.) As do most of the other rebuild projects, we change the kernel SRPM and do not label it .centosx. This is because the kernel needs to be exactly the same name to allow 3rd party modules to function.

I would like to thank the upstream vendor for making the SRPMS available in the manner that they do. There are several other enterprise vendors who do not make their source as readily available. Their product is excellent (or we would not rebuild it as CentOS), as is their support. If you need the support services that they provide, we highly recommend their

enterprise product.

6. Does CentOS change the upstream Source RPMs?

No. CentOS' key mandate for our base and updates repositories is NOT extending or enhancing packages or features beyond those supplied by the upstream Source RPM's. CentOS strives intentionally to provide binary functionality for our users. CentOS does offer other (optional) repositories called extras, addons, contribs, and centosplus that do offer added functionality. There is a Wiki page about the various CentOS repositories and their purposes.

7. Is there a Contrib area?

Yes. Please see the Wiki page on Contributing to the CentOS project

8. What is CentOS's relationship with Red Hat®, Inc. or RHEL?

Red Hat curates the trademarks for CentOS and is providing initial guidance and expertise required in establishing the formal board structure used to govern the CentOS Project.

Some members on the CentOS Project Governing Board work for Red Hat, Inc.

CentOS Linux is NOT supported in any way by Red Hat®, Inc.

CentOS Linux is NOT Red Hat® Linux, it is NOT Fedora™ Linux. It is NOT Red Hat® Enterprise Linux. It is NOT RHEL. CentOS Linux does NOT contain Red Hat® Linux, Fedora™, or Red Hat® Enterprise Linux.

CentOS Linux is **NOT** a clone of Red Hat® Enterprise Linux.

CentOS Linux is built from publicly available source code provided by Red Hat®, Inc for Red Hat® Enterprise Linux in a completely different (CentOS Project maintained) build system.

9. Where can I get package XyZ.rpm for CentOS?

The Wiki has a page about the CentOS and other "friendly" repositories. Chances are good that one of those repositories has the package you are looking for.

10. What architectures are supported?

Only Major versions still in support upstream are considered here. Previously other architectures had candidates as well, but this is the current list as of January 2019:

CentOS 6 currently supports x86 and x86_64.

CentOS 7 currently supports x86_64. The AltArch Special Interest Group has community maintained (non-official) versions of CentOS 7 for i686, armhfp (Arm32), aarch64 (Arm64), ppc64le, and ppc64.

11. The upstream provider offers Enterprise Linux in several flavors, AS, ES, WS, PWS, etc. Which one is CentOS like?

CentOS is built from the publicly provided AS Enterprise Sources, although all of the above versions are built from the same sources. AS is either a larger subset of packages (than PWS and WS) or has advanced Kernel parameters supporting larger number of processors or memory (as compared to ES).

With the upstream provider, AS supports some IBM architectures not supported by the other versions (ES, PWS, WS). CentOS is built like the AS version.

Consult: What release am I running?

12. How can I easily compare what major package versions are in each version of CentOS?

On the CentOS Distro Page at DistroWatch.com you can compare Major packages and all tracked packages. DistroWatch is a good resource for comparing Linux and BSD distributions.

13. What are all the CentOS repositories (directories) and what is each one for?

See the Repositories page for more information.

14. How do I download and burn the CentOS isos?

You can download the latest CentOS ISOs from here: CentOS Downloads

After you download the ISOs, you should check the MD5 sums (or better sha1 or sha265 sums) of the ISO file(s) that you downloaded against the published md5sum (again sha1 or sha265 sums) list in the ISO directory. If the values match, the download is good ... if they do not match, the file was not downloaded correctly, and you need to get the file over again. Bittorrent downloads are best, because they do an MD5sum check as part of the download process. Generally the DVD media is preferable as it avoids switching media during installation, and can be used via the pre-defined [c7-media] repo definition for local installation of additional packages via yum. Some environments block P2P file sharing including Bittorrent, in which case direct download is the only viable option. The mirrors that support it can be found via the "Downloads/Mirrors" drop-down on the menu bar at the top of Home or Forum pages. Then pick the link Current CentOS Public Mirror List, pick your geographic region, and look down the column labeled "Direct DVD Downloads" to find a suitable mirror near you.

Once you have verified the checksums of the ISO, you know you have a good download. Now you can burn the ISO to a disk. If you have k3b (CentOS-4 users do, all other CentOS users do not by default) I recommend you use it. You want to use the Tools -> CD -> Burn CD Image or Tools -> DVD -> Burn DVD ISO Image option to write the ISO file to a CD/DVD.

More information on how to verify and burn a CD ISO image can be found at http://www.centos.org/docs/5/html/CD_burning_howto.html.

Once the CD is burned, you should be able to boot from it. The last check you need to do is to verify the media. This will verify that the writing of the ISO to your media happened correctly. There will be a Check Media option after you select your keyboard and language.

If your media passes this check (make sure to check each disc for multiple media sets), you have a fully working installable media. If it fails this check, but passed the md5 (or sha1 or sha265 sum) check above, then the problem is with the burned media. Try burning on new media at a slower speed, if possible.

All CentOS ISOs that we release have been checked, so if the checksums that you have match, the ISOs should burn clean and pass the media checks. If they do not, the problem is almost always a bad media write to CD/DVD.

If you would rather buy your CentOS ISOs already burned, please see our official CentOS CD/DVD Vendors page. These official CentOS vendors donate a portion of each CD/DVD sale directly to the CentOS Project. You get a tested ISO ready to use ... we get money ... does it get any better than that

15. Can I add X to my server after install? Or I installed from the Server ISO and it didn't install X, how do I install X?

The easiest way to install X (and a GUI system) is to use the `yum groupinstall` feature. First you can see all the yum groups available with the command: yum grouplist You can install X and Gnome or KDE as follows:

CentOS-5 yum groupinstall "X Window System" "GNOME Desktop Environment" or yum groupinstall "X Window System" "KDE (K Desktop Environment)"

CentOS-6 yum groupinstall "X Window System" "Desktop" or `yum groupinstall "X Window System" "KDE Desktop".

You may also want to add some other groups from the list like `"Graphical Internet" or "Office/Productivity"`

16. How do I create a xorg.conf file?

Upstream X developers have worked to make the **/etc/X11/xorg.conf** un-needed in the usual case. However sometimes one needs to 'tweak' settings for multiple display heads or such. A template **xorg.conf** may be created by root thus: {{{Xorg :1 -configure }}} which produces a file named: **xorg.conf.new** in the current working directory. This file may be tested using the **-config** option for **Xorg**. Once suitable, a backup may be made and the file can be placed in the **/etc/X11/** directory to be used as the default configuration file.

17. What are the Maximum number CPUs, Maximum filesystem size, Minimum / Maximum Memory and other specifications associated with the different CentOS versions?

See the CentOS Product Wiki Page

18. There is no mp3 support in CentOS! Or is there? How about other multimedia formats?

Yes. There is no mp3 support in CentOS, as there is no mp3 support in the distribution sources as provided from upstream. While the CentOS Project simply rebuilds what is available upstream, these have been excluded because of legal (patent license) issues. You either can encode your music files to ogg vorbis, which is supported in CentOS or install mp3 support from a third party repository, such as rpmforge. For example: If you want mp3 support for xmms, then install **xmms-mp3** from that repository.

The same is true for several other multimedia formats (codecs, for example: gstreamer plugins) and multimedia players like xine or mplayer. These aren't packaged with CentOS because of legal issues, but you may find those in the rpmforge repository.

19. What is the support "end of life" for each CentOS release?

CentOS 3	CentOS-3 updates until October 31, 2010
CentOS 4	CentOS-4 updates until February 29, 2012
CentOS 5	CentOS-5 updates until March 31, 2017
CentOS 6	CentOS-6 updates until November 30, 2020
CentOS 7	CentOS-7 updates until June 30, 2024

20. Where can I get the latest version of XyZ.rpm for CentOS? I cannot find it anywhere.

CentOS is an Enterprise-class operating system and as such is more about stability and long-term support than cutting edge. Major package versions are retained throughout the life cycle of the product. This is generally what Enterprise wants and affords developers a stable base on which to develop without fear that bespoke applications will break every time something gets upgraded to the latest and greatest, but ultimately buggy version or the API changes breaking backwards compatibility.

So no, you will generally NOT find the very latest versions of various packages included in an Enterprise-class operating system such as CentOS. It's a feature not a deficiency.

21. A PCI audit says I am running a version which has CVE exploits in it

It is of course possible if you have not run updates, or have not rebooted after an update. As always in a maintenance strategy, you should test updates on a non-production machine, and have a current and tested backup, taken **before** running the update tool.

Security patches and bug fixes are backported into the shipped version. See here for details: https://access.redhat.com/security/updates/backporting . Simply reading a version number on a package or a banner from network scanning is not sufficient to indicate a vulnerability, in light of this approach. Most reputable vendors understand this, but some seem to not account for the upstream approach in their product's reporting interface.

The **changelog** of each package generally specifies CVE matter addressed with patches. A CVE number is a commonly used reference to refer to vulnerabilities. As an example consider **httpd** which may be examined thus: {{{rpm -q --changelog httpd | less }}} or even this: {{{rpm -q --changelog httpd | grep CVE }}}

People regularly appear in CentOS support venues asserting that a PCI assessor, or a web driven scanning tool is reporting that they need to update to some specific version of software not shipped in CentOS. PCI does not mandate specific version levels, but rather freedom from known testable vulnerabilities, usually expressed in the shorthand CVE number fashion. If

a scan report is complaining about package versions, the person providing it is probably **not doing it right**, as the popular meme goes. CentOS and its upstream are continuously updated, and the CVE's addressed are reflected in the aforementioned changelog, so running a protective backup, updating, and rebooting or restarting the affected daemon service should address the matter. Other approaches, such as using one keyed to package version numbers, are simply wrong.

Please also read about the Wiki article Software Installation from Source. Obviously in the case where the administrators of a given installation have undertaken to extend a CentOS installation with local or non-CentOS provided binaries, the CentOS project cannot provide updates or maintain such divergence.

22. How do I install or update on a system with no network connection?

See the Wiki article Creating Update Media. The technique described there also works with DVD installation media. The key technique is

```
yum --disablerepo=\* --enablerepo=c5-media <yum_command_and_argument(s)>
```

23. I installed the x86_64 version, so why do I have i386 packages, and can I get rid of them?

CentOS follows the upstream source in this respect, as it does in general, and the x86_64 installation by default will install iX86 32-bit packages on a 64-bit installation for compatibility purposes. Many server system administrators (and some desktop users) want a pure 64-bit system and so remove all 32-bit packages. This can be accomplished as follows:

```
yum remove \*.i\?86
```

To keep any 32-bit packages from being installed in future updates, edit your /etc/yum.conf and add the line:

```
exclude = *.i?86
```

Be aware that 32-bit applications, including some third-party (non-CentOS) browser plugins that may only be available in 32-bit versions, will no longer work after this procedure.

You may also want to do this:

```
yum reinstall \*
```

The reason is that sometimes the **/usr/share/** items (shared between BOTH packages) get removed when removing the 32-bit RPM packages.

24. How do I provide appropriate information about my system when asking questions?

See How to Provide Information About Your System.

25. How do I upgrade from one major release to another?

Upgrades in place are not supported nor recommended by CentOS or TUV. A backup followed by a fresh install is the only recommended upgrade path. See the Migration Guide for more information.

26. Why was there no package selection step when I've tried to install CentOS using the LiveCD/LiveDVD?

Installing CentOS from the live images is just a simple transfer of the image that exists already on the CD (or DVD). Once the image is copied to the hard disk, you can adjust the set of installed packages using yum as you would do on any other CentOS system. If you need to make the selection of packages at install time, please use one of the other installation disks instead of the live isos.

27. Why is the CentOS Project Spamming me?

They probably are not doing so. CentOS is an Operating System, like Microsoft Windows 7 or Mac OSX. Operating systems are installed on computers and people then use those computers to do things. Some things they do are good, other things they do are bad. However, the CentOS Project has no control over what people do with computers where the CentOS operating system is installed, any more than Microsoft has any control over who you send email to if your computer has the Windows 7 operating system.

If you saw a page that looks like this and that is what lead you to believe that the CentOS Project sent you an email, then you need to read the bottom paragraph more carefully. It says this:

```
CentOS is an Operating System and it is used to power this website; however, the webserver is owned by the domain owner
and not the CentOS Project. If you have issues with the content of this site, contact the owner of the domain,
not the CentOS project.

Unless this server is on the CentOS.org domain, the CentOS Project doesn't have anything to do with the content on this
webserver or any e-mails that directed you to this site.
```

For example, if the website is **www.example.com**, you would find the owner of the **www.example.com** machine using the following WHOIS server and searching for **www.example.com**.

If you have an ip address, you can use the American Registry for Internet Numbers to find the owner of the machine. Note: If the IP address is not in the United States, ARIN will tell you the proper Regional Internet Registry to use.

It is the owner of the machine, and not the people who make the operating system that is sending you e-mail.

28. Why is the CentOS Project taking over my Website?

They are not. CentOS is an Operating System based on the Linux kernel. CentOS powers millions of Web Servers on the Internet, in fact it is currently the 2nd most used Linux operating system for web servers in the world (see Survey).

The "Powered By" page you are seeing is the default page when there is no content to show for a given site. This can be caused by a several issues, but the three most common are:

- You have just finished a web server install and you have not added any content to the machine.
- A web server misconfiguration where the configured location is empty.
- A Domain Name System issue where the name for the website in question is not pointing to the proper IP address.

The way to get this problem fixed is to talk to the person who configures either the web server or your domain names. The CentOS Project can not do either of those things on anyone else's machines.

29. How does CentOS versioning work?

1. CentOS Linux currently has 3 *major* released branches that are active: CentOS-6, CentOS-7 & CentOS-8.
2. CentOS Linux releases *minor* (point in time) versions of our major branches. Two very important things about CentOS Linux branches are:
    1. The CentOS Project provides updates or other changes **ONLY** for the latest version of each major branch. Thus, if the latest minor version of CentOS-6 is version 6.6 then the CentOS Project only provides updated software for this minor version in the 6 branch. If you are using an older minor version than the latest in a given branch, then you are missing security and bugfix updates. Older minor versions are not supported in official support avenues. Occasionally individuals may be able to provide some ad-hoc assistance but you will usually be asked to update to the latest version before recieving more specific help.
        - ***Note:*** Any minor version is just a *snapshot* with previous updates, plus the latest batch of new upstream updates, rolled into a new [base] repo with an initially empty [updates] repo.
        - ***Tip:*** There is a CentOS Vault containing older CentOS trees. This vault is a picture of the older tree when it was removed from the main tree, and does not receive updates. It should only be used for reference.
    2. When setting up yum repositories on CentOS Linux you should **ONLY** use the single digit for the active branch, which corresponds to the CentOS Linux major branch. For example, http://mirror.centos.org/centos/6/ , or http://mirror.centos.org/centos/7/ . This is because we move all older minor branches to http://vault.centos.org/ . Remember from the prior bullet point, no updates are ever added to minor versions of CentOS Linux once in the vault.
3. Since minor versions of CentOS are *point in time* releases of a major branch, starting with CentOS-7, we are now using a date code in our minor versions. So you will see CentOS-7 (1406) or CentOS-7 (1503) as a version. This way anyone can know, from the release, when it happened. In the above examples, the minor versions 1406 means June 2014 and 1503 means March 2015. In older major branches of CentOS, such as CentOS-6, we numbered things differently. Those branches are numbered as 6.0, 6.1, 6.2, etc.
    1. You might be wondering why the change with CentOS-7.
        1. We are trying to make sure people understand they can **NOT** use older minor versions and still be secure. Therefore, a date in the minor version allows users to know with a glance when this minor version was created. If it is older than many months, there is likely a new version you should look for.
        2. As organizations move from individual servers having individual functionality to virtual machine farms and cloud / container implementations, the CentOS Project is now producing VM, cloud, and container images as well as installer ISOs. These images have dates in their name by design. We want users to easily be able to know what major branch and minor release are in these images, again at a glance. If your CentOS-7 images have 1505 (May 2015) or 20150501 (May 1st, 2015) in the name then they are based on the latest minor release that comes before this date ... in this case CentOS-7 (1503).
4. You can see which source code from Red Hat Enterprise Linux was used to create each minor version of CentOS Linux from the chart entitled "Archived Versions" from http://wiki.centos.org/Download

2023-09-11 07:22